# Systematic Collapse of the Accretion Disc Across the Supermassive Black Hole Population

Scott Hagen,[1★] Chris Done,[1] John D. Silverman,[2,3,4,5] Junyao Li,[6] Teng Liu,[7] Wenke Ren,[8,2] Johannes Buchner,[9] Andrea Merloni,[9] Tohru Nagao[10,11] and Mara Salvato[9]

[1]*Centre for Extragalactic Astronomy, Department of Physics, Durham University, South Road, Durham, DH1 3LE, UK*
[2]*Kavli Institute for the Physics and Mathematics of the Universe (Kavli IPMU, WPI), UTIAS, Tokyo Institutes for Advanced Study, University of Tokyo, Chiba, 277-8583, Japan*
[3]*Department of Astronomy, School of Science, The University of Tokyo, 7-3-1 Hongo, Bunkyo, Tokyo 113-0033, Japan*
[4]*Center for Data-Driven Discovery, Kavli IPMU (WPI), UTIAS, The University of Tokyo, Kashiwa, Chiba, 277-8583, Japan*
[5]*Center for Astrophysical Science, Department of Physics & Astronomy, John Hopkins University, Baltimore, MD 21218, USA*
[6]*Department of Astronomy, Univerity of Illinois at Urbana-Champaign, Urbana, IL 61801, USA*
[7]*Department of Astronomy, University of Science and Technology of China, Hefei 230026, China*
[8]*CAS Key Laboratory for Research in Galaxies and Cosmology, Department of Astronomy, University of Science and Technology of China, Hefei 230026, China*
[9]*Max-Planck-Institute für Extraterrestrische Physik, Giessenbachstrasse 1, 85748, Garching bei München, Germany*
[10]*Research Center for Space and Cosmic Evolution, Ehime University, 2-5 Bunkyo-cho, Matsuyama, Ehime, 790-8577, Japan*
[11]*Amanogawa Galaxy Astronomy Research Center, Kagoshima University, 1-21-35 Korimoto, Kagoshima 890-0065, Japan*

arXiv:2406.06674v2 [astro-ph.HE] 1 Oct 2024

Accepted XXX. Received YYY; in original form ZZZ

## ABSTRACT

The structure of the accretion flow onto supermassive black holes (SMBH) is not well understood. Standard disc models match to zeroth order in predicting substantial energy dissipation within optically-thick material producing a characteristic strong blue/UV continuum. However they fail at reproducing more detailed comparisons to the observed spectral shapes along with their observed variability. Based on stellar mass black holes within our galaxy, accretion discs should undergo a transition into an X-ray hot, radiatively inefficient flow, below a (mass scaled) luminosity of $\sim 0.02\,L_{\rm Edd}$. While this has been seen in limited samples of nearby low-luminosity active galactic nuclei (AGN) and few rare changing-look AGN, it is not at all clear whether this transition is present in the wider AGN population across cosmic time. A key issue is the difficulty in disentangling a change in spectral state from increased dust obscuration and/or host galaxy contamination, effectively drowning out the AGN emission. Here we use the new eROSITA eFEDS Survey to identify unobscured AGN from their X-ray emission, matched to excellent optical imaging from Subaru's Hyper Suprime-Cam; allowing the subtraction of the host galaxy contamination. The resulting, uncontaminated, AGN spectra reveal a smooth transition from a strongly disc dominated state in bright AGN, to the collapse of the disc into an inefficient X-ray plasma in the low luminosity AGN, with the transition occurring at $\sim 0.02\,L_{\rm Edd}$; revealing fundamental aspects of accretion physics in AGN.

**Key words:** accretion, accretion discs – black hole physics – galaxies: active

## 1 INTRODUCTION

Active galactic nuclei (AGN) are powered by accretion onto a supermassive black hole (SMBH), but the structure of this flow is not well understood. Simple estimates of the accretion structure are that it should form a geometrically thin, relatively cool, disc if it is dense enough to thermalise (Shakura & Sunyaev 1973), giving a disc temperature which increases with decreasing radius from the black hole. This predicts a spectral energy distribution (SED) consisting of a sum of black-body components, typically peaking in the UV/EUV for bright AGN. This matches to zeroth order to the strong blue/UV continuum seen in these sources, but more detailed comparisons fail (Antonucci et al. 1989; Lawrence 2018).

One issue is that AGN always show X-ray emission, with a non-thermal tail extending to high energies (Elvis et al. 1994; Lusso & Risaliti 2016), indicating that some of the accretion energy must be dissipated outside of the optically thick disc structure. The X-ray emission is highly variable on short time-scales, indicating it originates in a compact coronal structure (e.g Lawrence et al. 1987; Ponti et al. 2012) Compton scattering seed photons from the disc to higher energies. Additionally, even the bright blue/UV continuum does not match in detail to the disc models. There is a ubiquitous downturn in the UV which occurs before the expected disc peak (Laor & Davis 2014; Cai & Wang 2023), which appears to connect to an upturn in the X-ray spectra below $\sim 1$ keV (generally referred to as the soft X-ray excess). These two features can be joined together across the unobservable EUV region to form an additional warm Comptonising component (Laor et al. 1997; Porquet et al. 2004; Gierliński & Done 2004), which cannot be explained by a either the standard disc model nor the hot Comptonising corona.

★ E-mail: scott.hagen@durham.ac.uk

© 2024 The Authors

Whatever the nature of the EUV component, this should carry most of the accretion power. EUV radiation is strongly ionising, producing characteristic broad emission lines where it irradiates nearby material (e.g Wills et al. 1985; Rees et al. 1989; Peterson & Wandel 1999; Kaspi et al. 2005). Current optical surveys take advantage of this when identifying and classifying AGN, by first selecting on the presence of a blue continuum in the photometry, and then confirming with spectroscopic follow up of broad emission lines (e.g Richards et al. 2002). These optically selected AGN span a wide range in mass, and so any given luminosity bin will span a wide range in Eddington ratio. However, controlling for black hole mass shows a sharp drop in the number of objects with $\dot{m} \lesssim 0.01$ (e.g Trump et al. 2011; Mitchell et al. 2023), where $\dot{m} = \dot{M}/\dot{M}_{\rm Edd} = L/L_{\rm Edd}$ (i.e. with the accretion efficiency factor included). This has generally been attributed to selection effects such as an increase in dust obscuration or host galaxy contamination, effectively hiding the AGN emission (e.g Fabian et al. 2008; Vasudevan et al. 2013; Hickox & Alexander 2018).

However, an alternative scenario is a change in the accretion structure, where the disc evaporates into a hot X-ray plasma, as often seen in galactic black hole binaries (BHBs) (Done et al. 2007). These objects show strong thermal disc emission, now with temperatures peaking in the X-ray ($\sim 10^7$ K), but only for $\dot{m} \gtrsim 0.02$ (Maccarone 2003). Below this there is a dramatic spectral transition, from being dominated by soft (low energy) X-ray thermal emission, to being dominated by harder (high energy) Comptonised emission from significantly hotter $\sim 10^9$ K X-ray plasma (Maccarone 2003; Done et al. 2007). These hard spectra are best described by an alternative solution to the steady state accretion flow equations, one that is hot, optically thin and geometrically thick rather than the cool, optically thin, geometrically thin accretion discs. The best known of these alternative solutions is the Advection Dominated Accretion Flow (ADAF: Narayan & Yi 1995). More generally there is a series of solutions at low density where the ion temperature is much higher than the electron temperature. Such a two temperature flow can only be maintained below a critical density (hence mass accretion rate/luminosity) as more frequent collisions give efficient thermalisation between electrons and ions. The specific ADAF solution exists up to a maximum of $\dot{m} \sim 0.01 - 0.02$ (Yuan & Narayan 2014), which is where we generally see the state-transitions.

If we scale these transitions (or differing accretion states) up to AGN we expect to see strong UV/EUV emission above $\dot{m} \sim 0.02$ where the flow can be characterised by a disc-like solution. However, below $\dot{m} \sim 0.01$, we expect little EUV emission, as now the flow would instead be characterised by a hot X-ray plasma giving a much harder Comptonised spectrum peaking in the hard X-rays rather than the EUV. In this model, the sharp drop in broad-line AGN below $\dot{m} \sim 0.01$ (e.g. Steinhardt & Elvis 2010) is an intrinsic feature rather than a selection effect, as the lack of a strong EUV continuum would naturally lead to an insufficient number of ionising photons required to induce the broad emission lines.

The lack of a UV excess (i.e a "big blue bump") has been seen in nearby low-luminosity AGN (Ho 1999, 2008; Nemmen et al. 2014), and in limited samples at higher redshifts (Trump et al. 2011). More recently, a few, very rare, changing-state (or changing-look) AGN exhibit behaviour where the disappearance of their broad emission lines coincides with a dramatic drop in the EUV continuum Noda & Done (2018), indicative of a transition in accretion state. This is now being strengthened, with recent samples of changing-look AGN showing they tend to reside around this transition region $\dot{m} \sim 0.01$ (Panda & Śniegowska 2024). In addition, recent tidal disruption events appear to support a change in accretion structure dependent on the mass-accretion rate (e.g Wevers et al. 2021; Yao et al. 2022; Sfaradi et al. 2022) However, it has not been established whether this transition is present across the wider AGN population (Maoz 2007) and over cosmic time.

In this paper we use X-ray emission to identify AGN, as (unlike BHBs) they always show a significant high energy tail (Elvis et al. 1994; Lusso & Risaliti 2016), and so should allow us to select objects both below and above the $\dot{m} \sim 0.01$ transition (e.g Aird et al. 2018). We use the eROSITA eFEDS X-ray data (Liu et al. 2022) combined with optical imaging of the eFEDS field using Hyper Suprime-Cam (HSC) to select X-ray sources that are co-incident with galactic nuclei (Li et al. 2024), and whose X-rays are unabsorbed. Importantly, the excellent HSC multi-band (*grizy*) imaging, provided by the Subaru Strategic Program (Aihara et al. 2022) allows for detailed decomposition of the unresolved AGN emission from the host galaxy (Ishino et al. 2020; Li et al. 2021a, 2024), allowing us to confidently reconstruct optical emission from low luminosity AGN where the total emission is generally dominated by the host galaxy. We then stack the data from each source in bins of black hole mass and monochromatic 3500 Å luminosity (used as a proxy for $\dot{m}$), to obtain mean AGN SEDs within each bin. This allows us to confidently model the emission both below and above the $\dot{m} \sim 0.01$ transition, showing clearly a change in the accretion structure from a flow dominated by a disc-like continuum to one dominated by an X-ray plasma, as $\dot{m}$ decreases.

The paper is organised as follows. In section 2 we provide an overview of the sample used in this paper. Then in section 3 we give an oveview of our SED model, and apply it to the mean SEDs within each bin, showing conclusively the transition in accretion state. Finally in section 4 we give a discussion on our results along with our conclusions. Throughout we will assume a standard cosmology, from the Planck 2018 results (Planck Collaboration et al. 2020).

## 2 THE SAMPLE

### 2.1 Sample Definition

We base our study on the combined HSC-eROSITA eFEDS AGN sample described in Li et al. (2024) (HSC) and Liu et al. (2022) (eROSITA). Briefly, the eFEDS region has 22079 AGN identified from X-ray point sources with ancillary optical-mid IR data to allow photometric redshift estimations (Salvato et al. 2022). Our study requires a confident decomposition of the AGN from the host galaxy. This decomposition is most reliable in the redshift range $0.2 \leq z \leq 0.8$, and restricting the sample to this range results in 4975 X-ray AGN. These are cross-matched to the HSC catalogue using the optical counter-part positions from Salvato et al. (2022), giving 3796 X-ray AGN with excellent imaging data for confident host-AGN flux decomposition, described in Li et al. (2021a); Ding et al. (2021). As our study is focused on the nuclear emission originating from the AGN itself we select only sources that are X-ray unobscured ($N_H < 10^{22}\,{\rm cm}^{-2}$, using the values measured in Liu et al. (2022)), which reduces the sample to 3509 AGN (Li et al. 2024). Additionally, to ensure sufficient signal-to-noise in the X-ray detection we discard any sources with less than 10 counts, $N_\gamma$ (as seen by eROSITA).

The HSC decomposition separates the unresolved nuclear emission, modelled by the HSC PSF, from the extended stellar emission from the host galaxy modelled with a smooth analytic Sersic function using the 2D image decomposition software tool LENSTRONOMY (Birrer & Amara 2018; Ding et al. 2021), (see: Ding et al. 2021; Li et al. 2021a). For additional details on the decomposition methodology for our sample we refer the reader to Li et al. (2024). We also

refer the reader to Appendix A for details showing the effect of the decomposition on the SED.

In order to confidently asses the AGN flux for the low-luminosity AGN we only include sources where the AGN optical flux is *at least* 5 % of the host galaxy flux in all HSC bands, since below this the decomposition results become unreliable. Additionally we also discard sources with a poor fit to the HSC image (defined as a source with $\chi^2_\nu > 5$).

In summary, our selection critera can be written as follows:

- $F_{\rm AGN,HSC} \geq 0.05 F_{\rm Host,HSC}$
- HSC Image $\chi^2_\nu < 5$
- $N_{H,\rm X-ray} < 10^{22}\,{\rm cm}^{-2}$
- $N_{\gamma,\rm X-ray} \geq 10$

These selection criteria reduce our final sample to 2759 sources.

### 2.2 Mass Estimates

Only a small portion of our sample (275 sources) overlap with the SDSS DR 16 AGN catalogue, which provides mass estimates from broad-emission lines (Wu & Shen 2022). This is not necessarily surprising as we have intentionally used an X-ray selection that picks AGN regardless of the presence of a blue-continuum and/or broad emission lines such that we can test whether low-luminosity AGN have these features or whether they look like the changing-state AGN (Noda & Done 2018). However, we do still require black hole mass estimates for all our sources in order to estimate their $\dot{m}$. The HSC images have stellar mass estimates for the host galaxy (see Li et al. 2024), and so we use these to estimate the black hole mass from the local $M_* - M_{\rm BH}$ relation (Ding et al. 2020):

$$\log_{10}\left(\frac{M_{\rm BH,local}}{10^7}\right) = 0.27 + 0.98\log_{10}\left(\frac{M_*}{10^{10}}\right) \tag{1}$$

There is a redshift dependent offset to this relation, as shown in Li et al. (2021b), which we add to Eqn. (1) to give the following $M_* - M_{\rm BH}$ relation:

$$\log_{10} M_{\rm BH} = \log_{10} M_{\rm BH,local}(M_*) + \Delta M_{\rm BH}(z) \tag{2}$$

where

$$\Delta M_{\rm BH}(z) = \gamma \log_{10}(1+z) \quad \text{where } \gamma = 1.00 \pm 0.07$$

As a sanity check we compare our derived black hole masses to the SDSS measured masses, for the 275 sources where we have overlap. This is shown in Fig. 1. It is clear from this that our mass-estimates work as an acceptable proxy, with the majority lying within 0.5 dex of the SDSS measured values. Additionally, it is also clear that our mass estimates are most reliable for black hole masses in the range $\log_{10} M/M_\odot = 8.0 - 8.5$, which is where we will focus the main analysis as this is the range where we also have the largest number of sources (see section 2.5).

### 2.3 Calculating Rest Frame Luminosities

Our sample covers a range in redshift (0.2-0.8), and so we need to convert all the observed fluxes to rest-frame luminosities on a common wavelength/energy grid before we can stack to create mean SEDs. Here we give a brief description of our methodology, for both the HSC and eROSITA data.

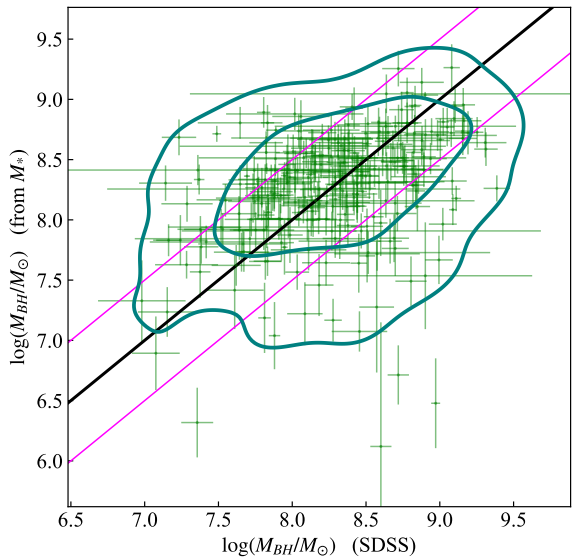


**Figure 1.** Black hole masses derived from the local $M_*$ relation (Ding et al. 2020; Li et al. 2021b) versus masses measured from single epoch SDSS spectra in the DR16 AGN catalogue (Wu & Shen 2022). The solid black line shows a 1:1 relation, while the magenta lines show a ±0.5 dex offset. The contours highlight the density distribution of the points, showing the $1\sigma$ and $2\sigma$ levels (inner and outer contours respectively).

#### 2.3.1 HSC

The HSC filters range from the g-band (~ 4000 Å–5500 Å) to the Y-band (~ 9000 Å–11000 Å). For the redshift range of our sample this corresponds to a rest frame wavelength range of ~ 3300 Å–6000 Å, to ensure all of our sources fall within this range. We define common rest-frame wavelength bins from 3350 Å to 5550 Å, each with a bin width of $\Delta\lambda = 550$ Å, as this gives evenly spaced bins that have a width close to the HSC filters yet are narrow enough to show changes in spectral shape.

For each source we first de-redden the observed frame AGN flux, using the extinction law of Fitzpatrick & Massa (2007), and $E(B-V)$ values from the dust maps of Schlafly & Finkbeiner (2011), stressing that this only corrects for the extinction within our own Galaxy. No attempt has been made to correct for extinction/reddening in the frame of the AGN.. We also include a cosmological K-correction on the flux-densities, by multiplying with a factor $(1+z)^{-1}$. We then convert the observed wavelength bins to rest frame wavelength; $\lambda_{em} = \lambda_{obs}/(1+z)$. For each of the common rest-frame bins, $i$, we then calculate the fractional overlap, $w$, with each of the HSC rest-frame bins, $j$. This is defined such that $w_j = 1$ implies that the common rest-frame bin is fully contained in the HSC rest-frame bin (since we are working in flux-density units), $w_j < 1$ implies the bins only partially overlap, and $w_j = 0$ implies no overlap between the common and original bin. These are then used to calculate the flux-density in the new common rest-frame bin, $F_{c,i}$, using a weighted log-mean:

$$\log_{10} F_{c,i} = \frac{\sum_j w_j \log_{10} F_{\rm HSC,j}}{\sum_j w_j} \tag{3}$$

where $F_{\rm HSC,j}$ is the HSC flux-density in the filter $j$ (where $j$ iterates